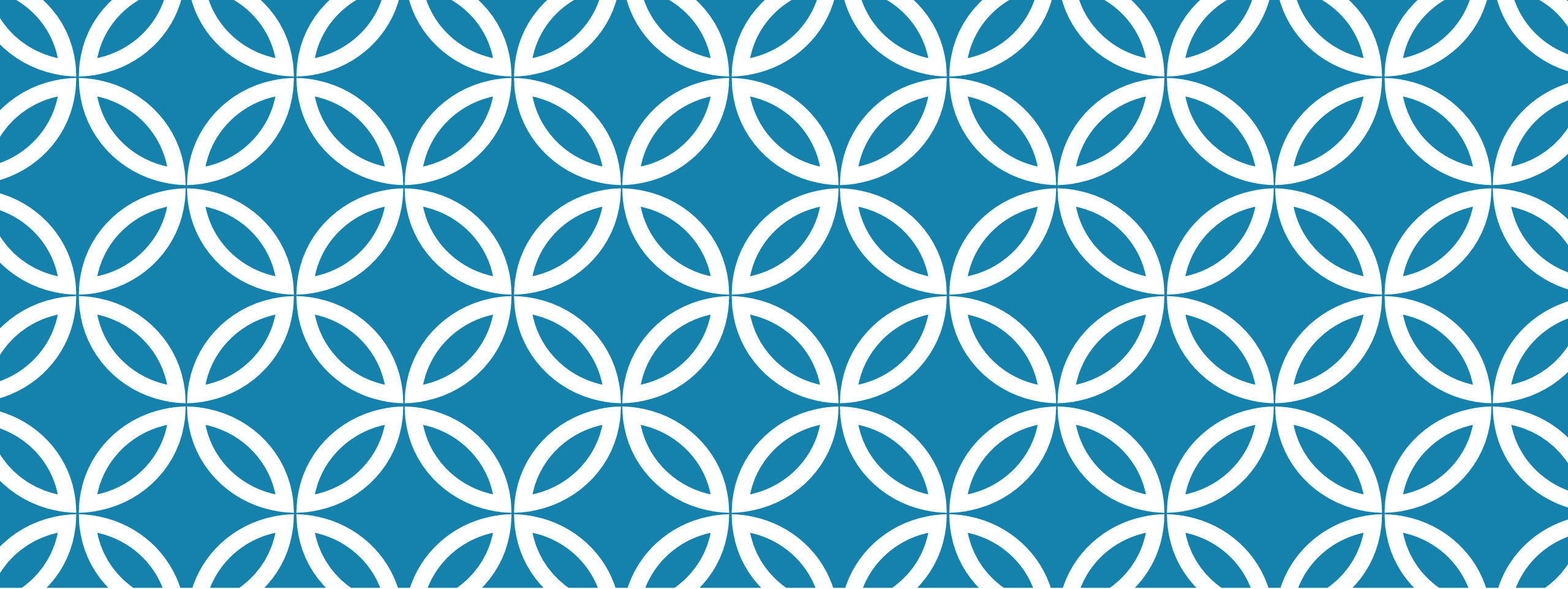




INTRODUCTION TO ESTIMATION

Lecture 1

WHAT IS AN ESTIMATE?

- An estimate may be defined as the process of calculating the quantities and costs of various items required in connection with the work.
- It is prepared by calculating the quantities from dimensions on the drawings for the various items required to complete the project and multiplying those quantities by the unit cost of the items concerned.
- To prepare an estimate, drawings consisting of the plan, elevation & sections through important points, along with a detailed specifications giving specific description of all workmanship, properties & proportion of materials are required.

PURPOSE OF ESTIMATING

- To ascertain the necessary amount of money required by the owner to complete the proposed work.
- For public construction works, estimates are required in order to obtain administrative approval, allotment of funds and technical sanction.
- To ascertain quantities of materials required in order to programme their timely procurement. To procure controlled materials like cement, steel etc. quantities of such materials are worked out from the estimate of the work & attached with the application for verification.
- To calculate the number of different categories of workers to be employed to complete the work within the scheduled time of completion.
- To assess the requirements of Tools, Plants & Equipment required to complete the work according to the programme.

PURPOSE OF ESTIMATING

- To fix up the completion period from the volume of the works involved in the estimate.
- To draw up a construction schedule & programme & also to arrange the funds required according to the programme.
- To justify the investment from the benefit-cost ratio (ratio > 1 is needed)
- To invite tenders & prepare bills for payment.
- An estimate for an existing property is required for valuation (calculation of market value for dealing real estate transactions)

WHY IS AN ESTIMATE NEVER THE ACTUAL COST?

- An estimate is the probable cost of construction for a work, worked out from the dimensions on the drawings & the standard rates at the time of preparation of the estimate.
 - The accuracy of the estimate depends upon –
 - How carefully the dimensions were done on the drawings.
 - The method of estimation used for calculation
- Rates & probable expenses which may be required for various unforeseen works & labourers.
- The actual cost of construction can be known from the physical measurements of the dimensions for various items of work at site and the actual rates of such items of work during execution of the work.
- So, all estimates are approximate in nature and effort should be made to keep the difference between the estimated cost and the actual cost to a minimum.

DIFFERENT TYPES OF ESTIMATES

ESTIMATES CAN BE OF THE FOLLOWING TYPES:

1. Detailed estimate
2. Preliminary /approximate/rough estimate
3. Quantity estimate or quantity survey
4. Revised estimate
5. Supplementary estimate
6. Combination of revised and supplementary estimate
7. Revised and supplementary estimates due to reduction in cost (following PWD manual)
8. Complete estimate
9. Annual maintenance or repair estimate

DIFFERENT TYPES OF ESTIMATES

1

A DETAILED ESTIMATE

- This includes the detailed particulars for the quantities, rates and costs of items involved for satisfactory completion of the project.
- Quantities of all items of work are calculated from their respective dimensions on the drawings on a MEASUREMENT SHEET
- Multiplying these quantities by their respective rates in a separate sheet, the cost of all items of work are worked out individually and then summarized i.e. ABSTRACTED (which is the detailed actual estimated cost of work).
- All other expenses required for satisfactory completion of the project are added to the above cost to frame the total of a detailed estimate.

DIFFERENT TYPES OF ESTIMATES

1

A DETAILED ESTIMATE

- A detailed estimate is accompanied by :-
 - Report
 - Specifications
 - Detailed drawings showing plans, different sections, key plan.
 - Design data and calculations
 - Basis of rates adopted in the estimate.
- A detailed estimate is prepared for technical sanctions, administrative approval, inviting tenders and for execution of a contract with the contractor.

DIFFERENT TYPES OF ESTIMATES

2

A PRELIMINARY /APPROXIMATE/ROUGH ESTIMATE

- This is a calculation of the approximate cost in a short time and enables the authority concerned to consider the financial aspect of the scheme.
- Such an estimate is framed after knowing the rate of similar works and from practical knowledge. Some popular methods are:-
 - Plinth area / square meter method
 - Cubic rate method
 - Service unit / unit rate method
 - Bay method
 - Approximate quantities with bill method
 - Cost comparison method
 - Cost of materials and labour.

DIFFERENT TYPES OF ESTIMATES

3

A QUANTITY ESTIMATE OR QUANTITY SURVEY

- This is the complete estimate or list of quantities for all items of work required to complete the concerned project.
- The quantity of each item of work is measured from dimensions on the drawings.
- Then the quantity is multiplied with the rate per unit for that item to get the cost.
- The purpose of the bill of quantities is get a complete list of items necessary and when priced gives the estimated cost.

DIFFERENT TYPES OF ESTIMATES

4

REVISED ESTIMATE

- A revised estimate is a detailed estimate for the revised quantities and rates of items of work originally provided in the estimate.
- Without MATERIAL DEVIATIONS of a STRUCTURAL NATURE from the original design.
- It is accompanied with a COMPARATIVE STATEMENT ABSTRACT FORM showing the probable variations for quantity rate and amount for each item of work of the project as compared with the original estimate side by side, stating the reasons of variation.
- A revised estimate is prepared and submitted for fresh technical sanction.

DIFFERENT TYPES OF ESTIMATES

4

REVISED ESTIMATE

A revised estimate may be required for the following reasons:

- When a sanctioned estimate is likely to exceed by more than 5% either from the rates being found insufficient or from any cause other than important structural alterations.
- When the expenditure of work exceeds or is likely to exceed by more than 10% of the administrative approval (for works worth more than Rs. 5 Lakhs).
- When there are material deviations (e.g. specification changes) from the original proposal but not due to material deviation of a structural nature.
- If it is found that the sanctioned estimate is more than the actual requirements.

DIFFERENT TYPES OF ESTIMATES

5

A SUPPLEMENTARY ESTIMATE

- While a work is in progress, some changes or additional works due to material deviations of a structural nature from the original design may take place.
- An estimate is then prepared to include all such works. This is known as a supplementary estimate.
- The method of preparation of a supplementary estimate is the same as that of a detailed estimate and it should be accompanied by a full report of the circumstances which render it necessary.
- The abstract must show the amount of the original estimate and the total of the sanction required including the supplementary amount.

DIFFERENT TYPES OF ESTIMATES

6

COMBINATION OF REVISED AND SUPPLEMENTARY ESTIMATE

- During execution of a project it may be necessary to revise the original estimate due to increased volume of original proposed work and at the same time sanction of supplementary works.
- For such a case a revised estimate is prepared for the increased volume of original work and a detailed estimate for the supplementary works not included in the original schedule.
- The amount of supplementary estimate is added to the revised estimate, showing the amount separately.
- Thus the total amount of revised estimate includes the amount for supplementary works also.

DIFFERENT TYPES OF ESTIMATES

DIFFERENCE BETWEEN REVISED AND SUPPLEMENTARY ESTIMATE

REVISED ESTIMATE	SUPPLEMENTARY ESTIMATE
This is required when the sanctioned amount is exceeded due to change of rates or addition of works (dependent on original work).	This is required due to supplementary works which is independent of the work first sanctioned
Due to material deviation (non-structural kind) from the original proposed work	Due to material deviation (structural kind) from original design
It is accompanied with a comparative statement abstract form showing the probable variations of each item of work	No comparative abstract is required
No revision drawings are required	Revision drawings are required

DIFFERENT TYPES OF ESTIMATES

7

REVISED AND SUPPLEMENTARY ESTIMATES DUE TO REDUCTION IN COST (FOLLOWING PWD MANUAL)

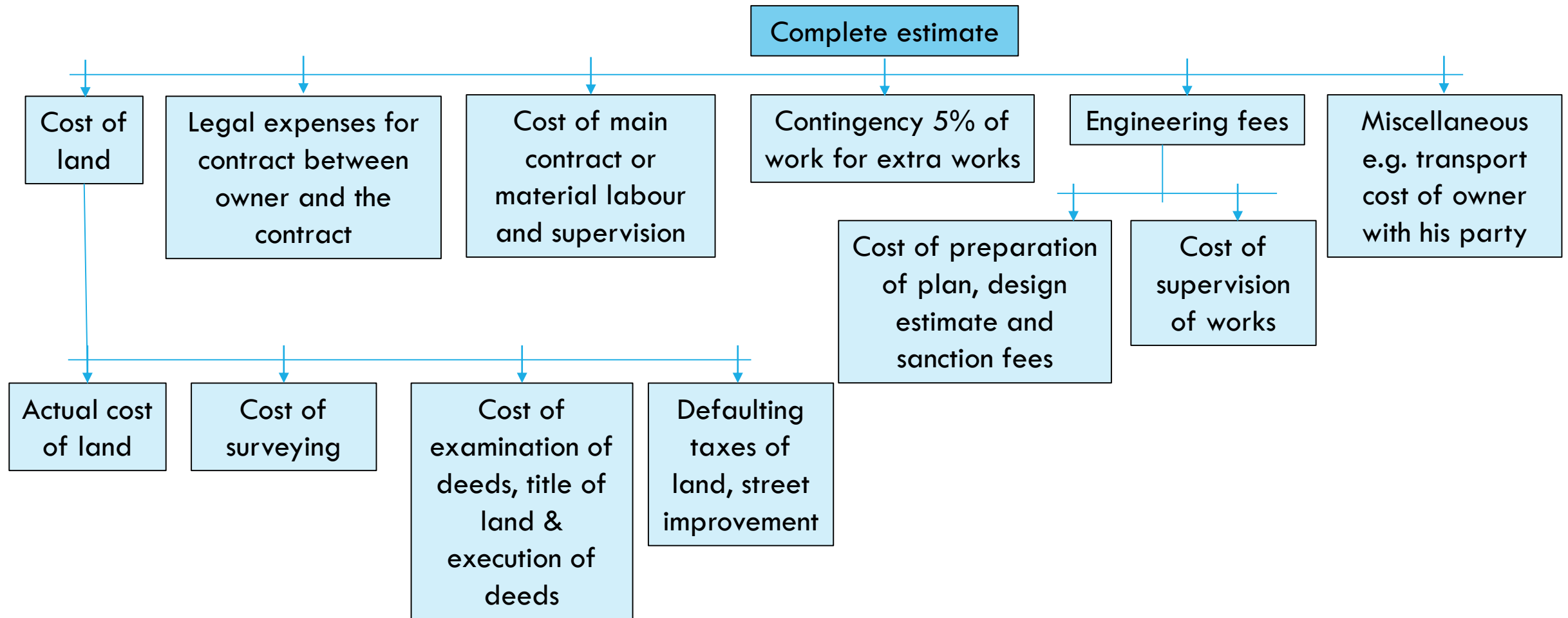
- Where a substantial section of a project (costing not less than 5% of the total sanctioned cost of the whole project) is abandoned or
- Material deviations from the original proposal is expected to result in substantial savings, the estimate is revised.
- But in cases where the saving is due to a material deviation of a structural nature from the design originally approved, supplementary estimate is prepared for the revised technical sanction.

8

DIFFERENT TYPES OF ESTIMATES

A COMPLETE ESTIMATE

- This is an estimated cost of all items which are related to the work in addition to the main contract or to the detailed estimate.



DIFFERENT TYPES OF ESTIMATES

9

ANNUAL MAINTENANCE OR REPAIR ESTIMATE

- After completion of work, it is necessary to maintain the same for its proper functioning and for the same an estimate is prepared for its items which require renewal, replacement, repairs etc. in the form of a detailed estimate.
- E.g. white washing, colour washing and painting of doors & windows etc. quantities are based on the previous measurements recorded in the measurement books.
- For petty works such as replacement of glass panes, repair of floors, path repair to cement plaster etc. which cannot be estimated in the beginning – only LUMPSUM PROVISION is made for different items in the estimate. Approximate idea is taken from previous year's expenditure.